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$$\begin{aligned}\lim_{x \rightarrow 1} \left(\frac{\sqrt{x} - x}{x - 1} \right) &= \frac{0}{0} \text{ (onbepaald)} \\&= \lim_{x \rightarrow 1} \left(\frac{-\sqrt{x}(-1 + \sqrt{x})}{x - 1} \right) = \lim_{x \rightarrow 1} \left(\frac{-\sqrt{x}(\sqrt{x} - 1)}{(\sqrt{x} + 1)(\sqrt{x} - 1)} \right) \\&= \lim_{x \rightarrow 1} \left(\frac{-\sqrt{x}}{\sqrt{x} + 1} \right) = \frac{-1}{2}\end{aligned}$$

References